# Razan Alsulieman at StanceEval-2026: Validation Under Target Shift for Arabic Stance Detection

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## Abstract

We describe our systems for both StanceEval-2026 Arabic stance-detection tracks. Our MARBERT-centered pipeline optimized the primary macro-F1 over AGAINST and FAVOR directly, combining target-aware fine-tuning, probability ensembles, and guarded transfer experiments. We report one official test submission per track. The Track 1 target-routed MARBERT system scored 0.8925 on development and 0.6743 on the official test set. The Track 2 MARBERT/Fanar/Qwen3 system scored 0.8360 on development and 0.8100 on the official test set. The released Track 1 evaluation input contained only Women Driving, absent from its labeled train and development targets, making the nominal seen-target track an effective target-transfer test for our system. These results show that validation protocols must reflect the intended target-transfer setting; otherwise, same-target development scores can substantially overstate transfer performance.

**Keywords:** Arabic stance detection; cross-target generalization; target-shift validation; MARBERT

## 1 Introduction

Stance assigns a position toward a specified target, so target relevance and polarity must be resolved jointly. This differs from sentiment: negative language can support a target by attacking its opponents. The distinction is especially consequential for social-media Arabic, where dialect, sarcasm, and implicit reference coexist with limited target coverage. Mawqif introduced Arabic target-specific stance data with auxiliary sentiment and sarcasm labels (Alturayeif et al., 2022); StanceEval subsequently made seen- and unseen-target transfer explicit evaluation settings (Alsulieman et al., 2026a).

Our research question is: *which internal validation protocol predicts performance after an Arabic stance target changes?* This question arose from the sharp Track 1 transfer gap (0.8925 development to 0.6743 test), whereas Track 2's held-target validation tracked evaluation more closely (0.8360 to 0.8100). Rather than treating the remaining experiments as an unstructured sweep, we organize them around three hypotheses: metric-aligned variance reduction, explicit text–target relation modeling, and target-shift robustness. We contribute (i) a MARBERT-centered system optimizing FAVG2 directly; (ii) a comparison between same-target and held-target validation; and (iii) quantitative error and negative-result analyses showing what remained after broad intervention.

## 2 Transfer-Oriented Background

Cross-target stance is relational: lexical polarity alone does not determine a post's stance toward a target. Generalized topic representations and Arabic social-media pretraining motivate our pair and target-dropout variants (Allaway and McKeown, 2020; Abdul-Mageed et al., 2021). We also test variance reduction, robust optimization, and target/test-time adaptation (Wortsman et al., 2022; Jiang et al., 2020; Sagawa et al., 2020; Foret et al., 2021; Gururangan et al., 2020; Ganin et al., 2016; Wang et al., 2021; Iwasawa and Matsuo, 2021).

## 3 Task and Data

Each example contains an Arabic post and target and receives one label: AGAINST, FAVOR, or NONE. The primary metric is FAVG2 $= (F1_A + F1_F)/2$; NONE is excluded from the average but still changes scored-class precision and recall. We therefore selected checkpoints, ensembles, and decision rules using FAVG2, not accuracy.

Track 1 contains 3,502 training and 619 development examples over Women Empowerment, COVID Vaccine, and Digital Transformation; evaluation contains 352 Women Driving inputs.



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Track 2 has 2,721 labeled examples from COVID Vaccine and Digital Transformation, 1,400 held-out Women Empowerment development examples, and 644 evaluation inputs from E-cars (332) and Trimester (312). The benchmark extends Mawqif (Alturayeif et al., 2022) with held-out cross-target data (Albalawi et al., 2026b). No evaluation labels were available to our code.

Women Driving is related to Women Empowerment but was absent from labeled data and concerns a specific policy debate. Track 1 was thus an effective related-domain transfer test for our system, consistent with the task design (Albalawi et al., 2026a).

## 4 System and Experimental Logic

### 4.1 Metric-aligned MARBERT core

Our principal encoder is MARBERT, pretrained for diverse Arabic social media (Abdul-Mageed et al., 2021). Inputs concatenate the target and post. The core pool covered seeds {5, 7, 13, 21, 29, 37, 42, 88, 111}, learning rates $\{2, 2.5, 3\} \times 10^{-5}$, maximum lengths {128, 160, 192}, pair and concatenated inputs, and target-dropout rates {0, 0.10, 0.20}; this was a designed pool rather than a full Cartesian grid. Runs used four epochs. The specialist runs used train/evaluation batches of 8/32 and two-step gradient accumulation. We stored class probabilities, then searched weighted soft votes, target-specific weights, class multipliers, and explicit decision thresholds against FAVG2.

The Track 1 development record (0.8925) was a single routed probability vector built in two stages. For each known target, a target-specific search combined retained base-run probabilities with six newly trained pair-input MARBERT specialists: cross-entropy and weighted cross-entropy at seeds 111/37, focal loss ($\gamma = 1$) at seed 111, and label smoothing 0.05 at seed 37. All used learning rate $2 \times 10^{-5}$; seed 111 used length 128 and seed 37 used length 160. The three optimized target outputs were restored to the original row order; the global stage retained this assembled vector, and no later decision rule improved it.

The Track 2 MARBERT reference (0.8317) searched 5,488,640 weight vectors over a 64-file pool. Its six dominant members held 99.93% of the weight: seed 5, $2 \times 10^{-5}$/160/dropout 0.20 (0.255101); seed 29, $2 \times 10^{-5}$/192/dropout 0.20 (0.189896); seed 111, $2 \times 10^{-5}$/192/pair robust preprocessing (0.171402); seed 111, $2 \times 10^{-5}$/192/baseline (0.159499); seed 29, $3 \times 10^{-5}$/192/dropout 0.20 (0.121389); and seed 7, $3 \times 10^{-5}$/160/baseline (0.102024). The remaining 43 active files shared 0.000689 weight. Multipliers were 1.633574, 1.634302, and 1.0 for Against, Favor, and None, respectively.

The reported Track 2 configuration first formed $b = .51p_M + .49p_F$ from the full MARBERT ($p_M$) and Fanar-1-9B-Instruct Q4_K_M ($p_F$) posteriors (Fanar Team et al., 2025). Let $\bar{b}$ and $\bar{q}$ denote the Against/Favor-normalized subvectors of $b$ and Qwen3-14B Q4_K_M output, respectively (Qwen Team, 2025). The two A/F-only blends were $r = .20\bar{b} + .80\bar{q}$ and $s = .30\bar{b} + .70r = .44\bar{b} + .56\bar{q}$. The final posterior was $((1 - b_N)s_A, (1 - b_N)s_F, b_N)$, thereby retaining the anchor's None probability. Qwen used one official-training demonstration per class, three samples, temperature 0.7, top-$p$ 0.8, top-$k$ 20, demonstration seed 20260724, and inference seed 20260731. This configuration scored 0.8360 on development and 0.8100 on the official test set.

### 4.2 Relation and robustness hypotheses

Relation variants included independent Against/Favor heads with a None gate, NLI (Yin et al., 2019), PET verbalizers (Schick and Schütze, 2021), multitask sentiment/sarcasm, and lexical hard pairs. Robustness variants covered contrastive learning, SMART, GroupDRO, soups, and SAM. Adaptation covered TAPT, gradient reversal, TENT, frozen T3A prototypes, prior correction, and conformal rules. Transductive methods used only unlabeled inputs. ArabicStanceX (Alkhathlan et al., 2025) and external LMs were isolated and disclosed.

### 4.3 Selection controls and implementation

Each run exported named class probabilities. Automated checks enforced row/ID order, normalized columns, computed class F1, and built one-label-per-line ZIPs. Manifests retained launch, environment, and decision-family metadata, allowing us to distinguish modeling changes from formatting errors and recompute tables.

Post-hoc search used a 121-point Against/Favor log-multiplier grid on $[-2.5, 2.5]$. Absolute thresholds used up to 121 empirical quantiles per class plus {0,.1,.2,1/3,.4,.5,.6,.7,.8,.9,1}, three tie rules (raw probability, normalized excess, and threshold ratio), and four None gates (stance sum, maximum stance probability, stance–None margin,

| Track | Development variant | FAVG2 | Δ | Interpretation |
|---|---|---|---|---|
| 1 | Six-model MARBERT ensemble | 0.8603 | — | Initial probability ensemble. |
| 1 | Target-routed specialists | 0.8925 | +0.0322 | Best development system. |
| 1 | Decoupled/PET/multitask | ≤0.8925 | ≤0 | None replaced the reference. |
| 2 | Best single MARBERT | 0.8106 | — | Seed 5, length 160, dropout 0.20. |
| 2 | Equal-weight top-eight | 0.8247 | +0.0141 | Variance reduction. |
| 2 | Optimized MARBERT vote | 0.8317 | +0.0070 | Weights and class multipliers. |
| 2 | Fanar full blend ($w = .49$) | 0.8359 | +0.0042 | External-model diagnostic. |
| 2 | Reported Fanar/Qwen blend | 0.8360 | +0.0001 | Final reported configuration. |

Table 1: Development ablation and variant comparison. Deltas are against the preceding row within each track; grouped alternatives are not cumulative.

and None rejection). We tested 131,902 Track 1 and 131,772 Track 2 rules without replacing scientific anchors.

For Track 1 evaluation, the development-selected Women Empowerment route was applied to Women Driving. Its two principal MARBERT members retained weights 0.51 and 0.38; an unavailable seed-88 member was replaced by the retained Women specialist at the same 0.11 weight. This fixed configuration was submitted as ID 857206 and scored 0.6743. Track 2 submission 866380 used the configuration specified above and scored 0.8100. Fanar and Qwen3 are disclosed external resources; neither reported system updated parameters using unlabeled evaluation inputs. Most runs used one Tesla P100 with PyTorch 2.1.2, CUDA 11.8, and Transformers 4.44.2.

## 5 Validation Protocol

Track 1 used the released same-target split. After its poor transfer, later candidates were also checked for prediction-distribution collapse; this cannot turn that split into Women Driving validation.

Track 2 used two-way leave-one-target-out (LOTO), reversing COVID Vaccine and Digital Transformation, with three paired seeds. Promotion required mean FAVG2 gain $\geq 0.0012$ and at least two positive seeds in each direction. The threshold came from a five-repeat null router range $[-0.000738, 0.000899]$: the expected five-draw normal range ($2.33\sigma$) implies single-run $\sigma \approx 0.0007$ and three-seed-mean $\sigma \approx 0.0004$, making 0.0012 an empirical $3\sigma$ noise margin, not a formal $p$-value. Total-variation and None-rate guards rejected collapse. Two source targets nevertheless make transfer estimates noisy.

| Track | Dev. | Test | Δ |
|---|---|---|---|
| Track 1 | 0.8925 | 0.6743 | -0.2182 |
| Track 2 | 0.8360 | 0.8100 | -0.0261 |

Table 2: Development and official test FAVG2 for one reported system per track. Deltas summarize the observed split-wise changes.

| Error family | Count | Share (%) |
|---|---|---|
| Against–Favor reversal | 175 | 72.0 |
| None boundary | 68 | 28.0 |

Table 3: Track 2 public-development errors for the 0.8317 anchor.

## 6 Results and Analysis

Track 1's $-0.2182$ gap reflects selection on known targets before testing on a new event-specific target. Track 2's gap was smaller ($-0.0261$), consistent with its held-target development design. Test transfer was uneven: 0.7704 on E-cars and 0.8393 on Trimester.

### 6.1 What errors remained?

On 1,400 Track 2 development rows, the 0.8317 anchor made 243 errors; 72% were direct A/F reversals (Table 3). Only 68 crossed None: 8 scored examples became None and 60 true None examples became scored. Low None F1 thus did not represent the largest error volume.

An unrealizable per-row oracle could recover 183/243 anchor errors (75.3%) and reach 95.7% accuracy. An A/F router failed out of fold, indicating useful pool diversity without enough source-target variation to learn its selection rule.

Three Track 1 development errors illustrate the boundary. ID 91 discussed digital-transition requirements without a stance (None→Favor). ID 541 paired apparent empowerment support with an absurd role reversal and religious ob-

jection (Against→Favor). ID 846 attacked anti-vaccine campaigners, so negative sentiment coexisted with Favor (Favor→Against). These pragmatic errors motivate explanation-aware models that organize argumentative evidence (Saha et al., 2024).

### 6.2 Negative evidence and external models

Decoupled heads, multitask learning, PET/NLI, and soups did not improve the anchors. An A/F router changed 1–3 predictions per repeat and averaged $+0.0000576$ (positive in 2/5); hard-pair mining failed its null-corrected gate. E-cars TAPT $(-0.003694, -0.001106)$ and Fanar prompt saturation $(-0.019665, -0.012486)$ were negative across both LOTO directions. DANN, TENT, T3A, SAM, XNLI, and conformal variants also retained the anchors.

Exact matching found 1,133/1,400 Track 2 development posts in LlamaLens's public Mawqif training split, which omits None. We excluded it from clean validation claims and from the reported systems. Fanar overlap remained unknown.

### 6.3 Why did the scores drop?

Track 1's 0.8925 measured interpolation over labeled targets, whereas evaluation introduced an event-specific target. Target-aware weighting amplified source regularities, but Women Empowerment did not supply Women Driving's policy framing or label boundary. The 0.2182 loss indicates optimistic validation, not a submission-format failure.

Track 2 asked the appropriate held-target question and differed by 0.0261, but two source targets could not teach target-conditioned polarity reversal. Oracle diversity existed; routing failed out of fold, and external signals were weak or difficult to certify. The bottleneck was selection, not another None threshold.

## 7 Conclusion

Same-target development selection overstated Track 1 transfer, whereas held-target validation tracked Track 2 more closely. Most residual errors reversed A/F, explaining why None calibration and larger correlated ensembles saturated. Future evaluations should align validation with the intended transfer setting.

## Limitations

Track 2 LOTO had only two labeled source targets, and hidden gold labels prevent test confusion or example analysis. Undisclosed external-model corpora make absent measured overlap inconclusive. Conclusions are specific to this Arabic dataset, metric, and model pool.


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